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Calculation and Uncertainty of Fast Radio Burst Structure Based on Smoothed Data*

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ABSTRACT

Studies of the time-domain structure of fast radio bursts (FRBs) require an accurate estimate of the FRB dispersion measure in order to recover the intrinsic burst shape. Furthermore, the exact DM is itself of interest when studying the time-evolution of the medium through which multiple bursts from repeating FRBs propagate. A commonly used approach to obtain the dispersion measure is to take the value that maximizes the FRB structure in the time domain. However, various authors use differing methods to obtain this structure parameter, and do not document the smoothing method used. Furthermore, there are no quantitative estimates of the error in this procedure in the FRB literature. In this letter, we present a smoothing filter based on the discrete cosine transform, and show that computing the structure parameter by summing the squares of the derivatives and taking the square root (that is, the 2-norm, $\sqrt{\Sigma(d/dt)^2}$) immediately lends itself to calculation of uncertainty of the structure parameter. We illustrate this with FRB181112 and FRB210117 data, which were detected by the Australian Square Kilometre Array Pathfinder, and for which high-time-resolution data is available.

Keywords: Radio transient sources (2008)—Astronomy data analysis (1858)

1. INTRODUCTION

Fast radio bursts (FRBs) are extragalactic radio transients of millisecond duration (Lorimer et al. 2007). Both the underlying mechanism of their emission, and the properties of the astrophysical plasmas through which they propagate, can be studied through detailed analysis of their time-frequency structure (e.g. Michilli et al. 2018; Macquart et al. 2020). The frequency dependent delay (dispersion measure, DM) of FRBs informs about the total integrated column density of free electrons — and hence total matter density — through which the FRB has propagated. Long-term studies of DM variation in repeating FRBs yield information on the medium in the vicinity of the progenitor (e.g. Zhao et al. 2021). Additionally, studies of FRB structure are sensitive to the assumed value of DM used to dedisperse the FRB (e.g. Hessels et al. 2019). To this end, FRB analysis aims to identify the correct DM, i.e. that corresponding to the integrated electron column density through which the FRB has passed. While FRB searches use the dispersion measure that maximizes signal-to-noise ratio (S/N), it is often assumed that the correct DM is that which maximizes structure (Caleb et al. 2020; Hilmarsson et al. 2021; Platts et al. 2021).

A common measure of structure is the time-derivative of the pulse intensity $I(t)$ (Gajjar et al. 2018; Hessels et al. 2019; Josephy et al. 2019; and B. C. Andersen et al. 2019; Pilia et al. 2020). However, since $I(t)$ is noisy, the derivative operation is not applied to a noisy estimate of the $I(t)$, but rather to a smoothed version thereof. This smoothing operation has the potential to affect the structure-maximizing DM, since over-smoothing will smear out the intrinsic FRB structure, while under-smoothing will leave the derivative dominated by noise. However, to date, there has been little attention paid on how to optimally select a smoothing time.

In the literature, there is also a slight variation in how the resulting d/dt are combined. For example, Hessels

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et al. (2019) used $\Sigma\left(d/dt\right)^2$, Gajjar et al. (2018) suggested $\Sigma\left|d/dt\right|$, and Josephy et al. (2019) experimented with $\Sigma\left(d/dt\right)^4$. One may argue that this is a matter of preference as long as the positive of the d/dt is combined and an interesting structure is detected. However, while any of these approaches should be able to identify the DM that maximises the structure in the signal, the uncertainty of the resultant fitted DM is also of considerable interest — without an accurate estimate of this uncertainty, it is impossible to determine whether, for instance, the DM of a given source is changing between bursts.

In §2, we shall demonstrate that the vector norm offers a meaningful measure of the uncertainty in the structure parameter inferred from noisy time-series data. In particular, the $\sqrt{\Sigma(d/dt)^2}$ (Euclidean or 2-norm) offers a physical and intuitive interpretation, as illustrated in §3. §4 then illustrates how the structure parameter, and uncertainty thereon, to be calculated for two example FRBs. Under the assumption that the "correct" DM of an FRB[1] is synonymous with structure maximization, this allows the correct DM to be identified, and assigned an uncertainty $\pm\Delta$DM which quantifies the confidence in such a detection.

2. METHOD

2.1. *Matrix representation and uncertainty definition*

Let $\mathbf{i} = [I(1), \cdots, I(N)]^T$ be a vector that contains discrete time samples of noisy total intensity $I(t)$. We may represent the smoothing operator as a matrix $\mathbf{S}$ and the first derivative as a matrix $\mathbf{D}_1$. Our immediate discussion here pertains to any $\mathbf{S}$ and $\mathbf{D}_1$ matrices, therefore the conclusions are generally applicable. It suffices to say, as we shall demonstrate later, it is indeed possible to write the smoothing algorithm as a matrix; for example, see Eilers (2003); Stickel (2010); Strang & Nguyen (1997). The result of the smoothing process is the smoothed estimate

$$\tilde{\mathbf{i}} = \mathbf{S}\mathbf{i}. \tag{1}$$

The first derivative operator can be expressed as a matrix, for example, a simple forward difference on N data points is a $(N-1)\times N$ matrix

$$\mathbf{D}_1 = \begin{bmatrix} -1 & 1 & & \\ & \ddots & \ddots & \\ & & -1 & 1 \end{bmatrix}. \tag{2}$$

[1] i.e. such that $\mathrm{DM} = \int n(e)(1+z)^{-1} d\ell$ for electron density $n(e)$ and unique signal propagation path $d\ell$.

Therefore, the first derivative of the smoothed data is

$$\mathbf{D}_1\tilde{\mathbf{i}} = \mathbf{D}_1\mathbf{S}\mathbf{i}. \tag{3}$$

The original noisy data can be expressed as the sum of the smoothed estimate and the difference between the noisy data and the smoothed version $\mathbf{\Delta_i} = \mathbf{i} - \tilde{\mathbf{i}}$.

$$\mathbf{i} = \tilde{\mathbf{i}} + \mathbf{\Delta_i}. \tag{4}$$

Eq. (4) splits the data into estimates of the signal and the noise. The true parts are $\mathbf{i} = \mathbf{s} + \mathbf{n}$, with $\mathbf{s}$ the noiseless burst and $\mathbf{n}$ zero-mean noise with covariance $\mathrm{Cov}(\mathbf{n}) = \langle \mathbf{n}\mathbf{n}^T \rangle$, where $\langle\cdot\rangle$ is the average over noise realisations. Since

$$\mathbf{\Delta_i} = (\mathbf{s} - \mathbf{S}\mathbf{s}) + (\mathbf{n} - \mathbf{S}\mathbf{n}), \tag{5}$$

$\mathbf{\Delta_i}$ contains the burst structure that smoothing removes, and $\tilde{\mathbf{i}} = \mathbf{S}\mathbf{s} + \mathbf{S}\mathbf{n}$ contains the in-band noise $\mathbf{S}\mathbf{n}$ that smoothing cannot remove. Therefore, Eq. (3) may be written as

$$\begin{aligned} \mathbf{D}_1\tilde{\mathbf{i}} &= \mathbf{D}_1\mathbf{S}(\tilde{\mathbf{i}} + \mathbf{\Delta_i}) \\ &= \mathbf{D}_1\mathbf{S}\tilde{\mathbf{i}} + \mathbf{D}_1\mathbf{S}\mathbf{\Delta_i}. \end{aligned} \tag{6}$$

Eq. (6) is illustrated in Fig. 1. We note that the first term in the right hand side of Eq. (6), $\mathbf{D}_1\mathbf{S}\tilde{\mathbf{i}} = \mathbf{D}_1\mathbf{S}\mathbf{S}\mathbf{i}$, is the derivative of the "double-smoothed" data. The second term, $\mathbf{D}_1\mathbf{S}\mathbf{\Delta_i}$, is the derivative of the smoothed "detrended" noise. By Eq. (5), however, the first term also contains in-band noise, and the second contains burst structure, $\mathbf{D}_1\mathbf{S}(\mathbf{s} - \mathbf{S}\mathbf{s})$.

The length (2-norm, $\|.\|_2$) of $\mathbf{D}_1\mathbf{S}\mathbf{\Delta_i}$ is the radius of the dashed circle in Fig. 1, which may be written as

$$\|\mathbf{D}_1\mathbf{S}\mathbf{\Delta_i}\|_2 = \sqrt{\sum_{n=1}^{N}\left[(\mathbf{D}_1\mathbf{S}\mathbf{i})_n - (\mathbf{D}_1\mathbf{S}\tilde{\mathbf{i}})_n\right]^2}, \tag{7}$$

where $(.)_n$ indicates the n-th element of the vector in question. The right hand side of Eq. (7) suggests that it can be interpreted as proportional to a standard deviation estimate ($\tilde{\sigma}\sqrt{N-1}$) of the first derivative of the smoothed data. Taking this as the uncertainty estimate of $\mathbf{D}_1\tilde{\mathbf{i}}$ means we take $\mathbf{D}_1\mathbf{S}\mathbf{\Delta_i}$ to represent a particular noise instance with standard deviation of $\|\mathbf{D}_1\mathbf{S}\mathbf{\Delta_i}\|_2/\sqrt{N-1}$. In that case, the tip of the vector $\mathbf{D}_1\tilde{\mathbf{i}}$ in Fig. 1 could lie anywhere on the circumference of the grey circle such that the estimate of relative uncertainty may be computed as

$$\frac{\|\mathbf{D}_1\mathbf{S}\mathbf{\Delta_i}\|_2}{\left\|\mathbf{D}_1\mathbf{S}\tilde{\mathbf{i}}\right\|_2}.$$

This radius is not the 1σ uncertainty of $\left\|\mathbf{D}_1\tilde{\mathbf{i}}\right\|$. Write $\mathbf{a} = \mathbf{D}_1\mathbf{S}\mathbf{s}$, $\mathbf{e} = \mathbf{D}_1\mathbf{S}\mathbf{n}$ and $\hat{\mathbf{a}} = \mathbf{a}/\|\mathbf{a}\|$, so that $\mathbf{D}_1\tilde{\mathbf{i}} = \mathbf{a} + \mathbf{e}$ exactly. Split the noise into its component along the structure, $e_\parallel = \hat{\mathbf{a}}^T\mathbf{e}$, and the rest, $\mathbf{e}_\perp = \mathbf{e} - e_\parallel\hat{\mathbf{a}}$. Then (Appendix B)

$$\left\|\mathbf{D}_1\tilde{\mathbf{i}}\right\| \simeq \|\mathbf{a}\| + e_\parallel + \frac{\|\mathbf{e}_\perp\|^2}{2\|\mathbf{a}\|}. \tag{8}$$

Only $e_\parallel$ changes the structure parameter at first order; the last term is a small positive bias. $e_\parallel = \hat{\mathbf{a}}^T\mathbf{D}_1\mathbf{S}\mathbf{n} = \mathbf{w}\mathbf{n}$ is a weighted sum of the noise samples, with the row of weights $\mathbf{w} = \hat{\mathbf{a}}^T\mathbf{D}_1\mathbf{S}$ fixed by the burst. Its mean is zero, and its standard deviation σ, the 1σ uncertainty, is given by

$$\sigma^2 = \mathbf{w}\,\mathrm{Cov}(\mathbf{n})\,\mathbf{w}^T, \tag{9}$$

or $\sigma^2 = \sum_t w_t^2\sigma_t^2$ for noise uncorrelated between samples with variance σ_t^2 at sample t. The relative uncertainty is

$$\frac{\sigma}{\left\|\mathbf{D}_1\tilde{\mathbf{i}}\right\|_2}. \tag{10}$$

The denominator is the structure parameter itself, $\left\|\mathbf{D}_1\tilde{\mathbf{i}}\right\| = \|\mathbf{D}_1\mathbf{S}\mathbf{i}\|$, the quantity whose scatter is σ; to first order it equals $\|\mathbf{a}\|$. The original denominator, $\left\|\mathbf{D}_1\mathbf{S}\tilde{\mathbf{i}}\right\| = \|\mathbf{D}_1\mathbf{S}\mathbf{S}\mathbf{i}\|$, treats the twice-smoothed data as the signal, as in Eq. (6). Its burst part is $\mathbf{D}_1\mathbf{S}\mathbf{S}\mathbf{s}$, not $\mathbf{a}$: per DCT mode the burst is weighted by $\mathrm{eig}_k^{1/2} f(k)^2$ rather than $\mathrm{eig}_k^{1/2} f(k)$. Since $f(k) \le 1$, and $f \simeq \frac{1}{2}$ near k_c, where the band pass of Fig. 2 peaks, $\|\mathbf{D}_1\mathbf{S}\mathbf{S}\mathbf{s}\| < \|\mathbf{a}\|$. In our simulations it is 5–6% low at any S/N, a fixed bias rather than noise, so it inflates the relative uncertainty by the same amount.

Eq. (7) instead measures the full length of the residual. By Eq. (5), $\mathbf{D}_1\mathbf{S}\boldsymbol{\Delta}_\mathbf{i} = \mathbf{d}_s + \mathbf{d}_n$ with $\mathbf{d}_s = \mathbf{D}_1\mathbf{S}(\mathbf{s} - \mathbf{S}\mathbf{s})$ and $\mathbf{d}_n = \mathbf{D}_1\mathbf{S}(\mathbf{n} - \mathbf{S}\mathbf{n})$, and (Appendix B)

$$\frac{\langle\|\mathbf{D}_1\mathbf{S}\boldsymbol{\Delta}_\mathbf{i}\|^2\rangle}{\sigma^2} = \frac{\langle\|\mathbf{d}_n\|^2\rangle}{\sigma^2}\left(1 + \frac{\|\mathbf{d}_s\|^2}{\langle\|\mathbf{d}_n\|^2\rangle}\right). \tag{11}$$

The structure parameter that is consistent with this uncertainty estimate is , evidently, the 2-norm of the $\mathbf{D}_1\tilde{\mathbf{i}}$ since first factor counts noise in every direction rather than only along $\hat{\mathbf{a}}$ (5–6 in our simulations). The second comes from the burst residual $\mathbf{d}_s$, which is not noise; $\|\mathbf{d}_s\|^2 / \langle\|\mathbf{d}_n\|^2\rangle$ grows as $(\mathrm{S/N})^2$ at fixed cutoff, and the factor is 4–13 in our simulations. Dividing Eq. (7) by $\sqrt{N-1}$ to get a per-sample standard deviation does not help either: the residual is concentrated near the filter cutoff k_c (§2.2), not white, and this underestimates σ by a factor of 5–10.

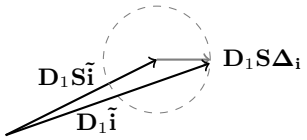


Figure 1. Vector diagram representing Eq. (6).

Eq. (6) and the triangle inequality give the bounds

$$\left\|\mathbf{D}_1\mathbf{S}\tilde{\mathbf{i}}\right\| - \|\mathbf{D}_1\mathbf{S}\boldsymbol{\Delta}_\mathbf{i}\| \le \left\|\mathbf{D}_1\tilde{\mathbf{i}}\right\| \le \left\|\mathbf{D}_1\mathbf{S}\tilde{\mathbf{i}}\right\| + \|\mathbf{D}_1\mathbf{S}\boldsymbol{\Delta}_\mathbf{i}\| \tag{12}$$

where the subscript $._2$ is not shown for brevity. Eq. (12) is valid for the 2-norm for N-dimensional vectors. It can be shown formally by applying the inner product definition $\mathbf{x}^T\mathbf{y} = \|\mathbf{x}\|_2\|\mathbf{y}\|_2\cos\theta$ in the expression $\|\mathbf{x}+\mathbf{y}\|_2^2 = (\mathbf{x}+\mathbf{y})^T(\mathbf{x}+\mathbf{y})$ (Strang 2019, see Ch. 1) and realizing that $-1 \le \cos\theta \le 1$ for real vectors.

In reality, the data is a vectorwith N entries. In that case, the circumference of the circle in These bounds are reached only when the residual is parallel to the structure vector. Only the parallel part of the noise, $e_\parallel$, counts at first order (Eq. 8), so a radius overstates the 1σ scatter (already by $\sqrt{2}$ for isotropic noise in the two-dimensional Fig. 1becomes the surface of an N-dimensional sphere with the same radius. The conclusion is unchanged); for the smoothed residual the factor is the square root of Eq. (11). The bound holds in N dimensions, but it is not a 1σ uncertainty.

Finally, the reciprocal of Eq. (10) is the structure to the structure noise ratio.

$$\mathrm{SSNR} = \frac{\left\|\mathbf{D}_1\mathbf{S}\tilde{\mathbf{i}}\right\|_2}{\|\mathbf{D}_1\mathbf{S}\boldsymbol{\Delta}_\mathbf{i}\|_2}.$$

$$\mathrm{SSNR} = \frac{\left\|\mathbf{D}_1\tilde{\mathbf{i}}\right\|_2}{\sigma}. \tag{13}$$

This quantity may offer a more prominent visual contrast in some cases.

For some applications, it may be desirable to test the hypothesis of whether the intrinsic structure in a signal $\mathbf{i}_1$ is significantly greater than that in another signal $\mathbf{i}_2$. This test asks whether or not the measured difference in structure,

$$\|\mathbf{D}_1\mathbf{S}\mathbf{i}_1\| - \|\mathbf{D}_1\mathbf{S}\mathbf{i}_2\|, \tag{14}$$

is significant compared to the relative noise,

$$\|\mathbf{D}_1\mathbf{S}\boldsymbol{\Delta}_{\mathbf{i}_1} - \mathbf{D}_1\mathbf{S}\boldsymbol{\Delta}_{\mathbf{i}_2}\| = \left\|\mathbf{D}_1\mathbf{S}\left(\boldsymbol{\Delta}_{\mathbf{i}_1} - \boldsymbol{\Delta}_{\mathbf{i}_2}\right)\right\|.$$

Continuing the analogy of Fig. 1, its noise. By Eq. (15) represents the surface of an n-dimensional sphere with magnitude given by a combination of $\mathbf{D}_1\mathbf{S}\boldsymbol{\Delta}_{\mathbf{i}_1}$ and $\mathbf{D}_1\mathbf{S}\boldsymbol{\Delta}_{\mathbf{i}_2}$. If the difference in structure between $\mathbf{i}_1$ and $\mathbf{i}_2$ given by (8), the noise in Eq. (14) is less than the radius of the sphere, the structures are consistent to within noise fluctuations. Specifically, the $e_{\parallel,1} - e_{\parallel,2} = \mathbf{w}_1\mathbf{n}_1 - \mathbf{w}_2\mathbf{n}_2$, with variance

$$\sigma_{12}^2 = \mathrm{Var}(\mathbf{w}_1\mathbf{n}_1 - \mathbf{w}_2\mathbf{n}_2). \tag{15}$$

The structure in $\mathbf{i}_1$ can be said to be greater than that in $\mathbf{i}_2$ when

$$\|\mathbf{D}_1\mathbf{S}\mathbf{i}_1\| > \|\mathbf{D}_1\mathbf{S}\mathbf{i}_2\| + \left\|\mathbf{D}_1\mathbf{S}\left(\boldsymbol{\Delta}_{\mathbf{i}_1} - \boldsymbol{\Delta}_{\mathbf{i}_2}\right)\right\|.$$

$$\|\mathbf{D}_1\mathbf{S}\mathbf{i}_1\| > \|\mathbf{D}_1\mathbf{S}\mathbf{i}_2\| + z\,\sigma_{12}. \tag{16}$$

Using $\left\|\mathbf{D}_1\mathbf{S}(\boldsymbol{\Delta}_{\mathbf{i}_1} - \boldsymbol{\Delta}_{\mathbf{i}_2})\right\|$ in place of σ_{12} is again a radius, and it contains the deterministic burst difference $\mathbf{D}_1\mathbf{S}[(\mathbf{s}_1 - \mathbf{S}\mathbf{s}_1) - (\mathbf{s}_2 - \mathbf{S}\mathbf{s}_2)]$, which grows away from the peak.

Such a test will become relevant when assessing whether the structure at one DM is significantly different from the structure at another, since $\boldsymbol{\Delta}_{\mathbf{i}_1}$ and $\boldsymbol{\Delta}_{\mathbf{i}_2}$ will in this case be correlated$\mathbf{n}_1$ and $\mathbf{n}_2$ are then the same channel noise with different delays. Near the peak $e_{\parallel,1} - e_{\parallel,2} \simeq e'_{\parallel}\,(d_1 - d_2)$ (Appendix B).

2.2. *Example smoothing matrix based on discrete cosine*

We can choose the eigenvectors of $\mathbf{D}_1^T\mathbf{D}_1$ as the basis for the smoothing operation. As we will show later, the close connection to the first derivative $\mathbf{D}_1$ matrix lends itself to insight when expressing the entire calculation as a single matrix operation. The eigenvectors of

$$\mathbf{D}_1^T\mathbf{D}_1 = \begin{bmatrix} -1 & 1 & & & \\ -1 & 2 & -1 & & \\ & . & . & . & \\ & & -1 & 2 & -1 \\ & & & -1 & 1 \end{bmatrix}, \tag{17}$$

are discrete cosines (Strang 1999). We shall call this eigenvector matrix $\mathbf{C}$ whose components are

$$\mathbf{c}_k(j) = \cos\left(j - \frac{1}{2}\right)\frac{(k-1)\pi}{N}, \tag{18}$$

where $k = 1, \cdots, N$ is the column number of $\mathbf{C}$ and $j = 1, \cdots, N$ is the row number. The eigenvalues are

$$\mathrm{eig}_k = 2 - 2\cos\frac{(k-1)\pi}{N}. \tag{19}$$

This is known as the discrete cosine transform-2 (DCT-2). Most importantly, DCT-2 has a fast implementation based on the fast Fourier transform (FFT) (Strang 1999). Therefore, numerical linear algebra eigendecomposition is not needed to obtain Eq. (18) and Eq. (19).

The smoothing operation may be written as

$$\tilde{\mathbf{i}} = \mathbf{C}\mathbf{D}_{\mathrm{filter}}\mathbf{C}^T\mathbf{i} = \mathbf{S}\mathbf{i}, \tag{20}$$

where $\mathbf{D}_{\mathrm{filter}}$ is a diagonal matrix representing a low-pass filter in the spectral domain; $\mathbf{C}^T\mathbf{i}$ is the DCT of $\mathbf{i}$ and $\mathbf{C}(\mathbf{D}_{\mathrm{filter}}\mathbf{C}^T\mathbf{i})$ is the inverse DCT of the filtered spectrum. The diagonal of $\mathbf{D}_{\mathrm{filter}}$ is the spectral response of a low-pass filter, for example,

$$f(k) = \frac{1}{1 + \left(\frac{k}{k_c}\right)^{2O}}, \tag{21}$$

where k_c is the spectral cutoff and O is the order of the filter. The resulting pass band is flat and the roll off rate is $2O$ orders of magnitude per decade, as inspired by the analog Butterworth filter. The DCT is purely real and does not force a periodic boundary condition (as in the case of FFT) that could introduce a discontinuity corresponding to high-frequency contents (Strang & Nguyen 1997, see Ch. 8).

Using smoothing process based on the DCT in Eq. (20), we can write

$$\begin{aligned} \left\|\mathbf{D}_1\tilde{\mathbf{i}}\right\|^2 &= \mathbf{i}^T\mathbf{C}\mathbf{D}_{\mathrm{filter}}\mathbf{C}^T\mathbf{D}_1^T\mathbf{D}_1\mathbf{C}\mathbf{D}_{\mathrm{filter}}\mathbf{C}^T\mathbf{i} \\ &= \mathbf{i}^T\mathbf{C}\mathbf{D}_{\mathrm{filter}}\boldsymbol{\Lambda}_{D_1^TD_1}\mathbf{D}_{\mathrm{filter}}\mathbf{C}^T\mathbf{i} \\ &= \left\|\boldsymbol{\Lambda}_{D_1^TD_1}^{1/2}\mathbf{D}_{\mathrm{filter}}\mathbf{C}^T\mathbf{i}\right\|^2 \end{aligned} \tag{22}$$

where we used the eigendecomposition $\mathbf{D}_1^T\mathbf{D}_1 = \mathbf{C}\boldsymbol{\Lambda}_{D_1^TD_1}\mathbf{C}^T$ and $\boldsymbol{\Lambda}_{D_1^TD_1}$ is a diagonal matrix of non-negative eigenvalues, eig_k, shown in Eq. (19); $\mathbf{C}^T\mathbf{C} = \mathbf{I}$ since $\mathbf{C}$ is an orthogonal matrix. Eq. (22) is a key result. It expresses the final product which is the sum of the square of the derivative of the smoothed data in terms of the end-to-end matrix operation. We note that $\boldsymbol{\Lambda}_{D_1^TD_1}^{1/2}$ is a diagonal matrix whose diagonal represents a high pass profile such that $\boldsymbol{\Lambda}_{D_1^TD_1}^{1/2}\mathbf{D}_{\mathrm{filter}}$ represents a band pass filter with the peak response at approximately k_c as shown in Fig. 2. The norm (subscript $_{.2}$ is not shown for brevity) of the smoothed data is the structure parameter, $\sqrt{\Sigma(d/dt)^2}$,

$$\left\|\mathbf{D}_1\tilde{\mathbf{i}}\right\| = \left\|\boldsymbol{\Lambda}_{D_1^TD_1}^{1/2}\mathbf{D}_{\mathrm{filter}}\mathbf{C}^T\left(\tilde{\mathbf{i}} + \boldsymbol{\Delta}_{\mathbf{i}}\right)\right\|. \tag{23}$$

Finally, the uncertainty residual length of Eq. (7) is

$$\|\mathbf{D}_1\mathbf{S}\boldsymbol{\Delta}_{\mathbf{i}}\| = \left\|\boldsymbol{\Lambda}_{D_1^TD_1}^{1/2}\mathbf{D}_{\mathrm{filter}}\mathbf{C}^T\boldsymbol{\Delta}_{\mathbf{i}}\right\|. \tag{24}$$

Expressing the 2-norm in the spectral domain as we have done above has an advantage in visualization. For example, $\mathbf{C}^T\mathbf{i}$ is the spectrum of $I(t)$. Pre-multiplying that quantity by $\mathbf{\Lambda}^{1/2}_{D_1^T D_1}\mathbf{D}_{\rm filter}$ filters the spectrum. Finally, taking the norm measures the length of the resulting vector.

In practice $\hat{\mathbf{a}}$ is estimated from the data, $\hat{\mathbf{a}} \simeq \mathbf{D}_1\mathbf{S}\mathbf{i}/\|\mathbf{D}_1\mathbf{S}\mathbf{i}\|$, and $\mathbf{w} = \hat{\mathbf{a}}^T\mathbf{D}_1\mathbf{S}$ is obtained by differencing $\hat{\mathbf{a}}$ and smoothing the result with $\mathbf{S}$. The noise level is measured from the same on-pulse spectrum in the band $k > N/4$, where the burst has no power,

$$\hat{\sigma}_n = 1.4826\,\mathrm{median}_{k>N/4}|(\mathbf{C}^T\mathbf{i})_k|, \qquad (25)$$

since the orthonormal DCT (columns of $\mathbf{C}$ normalised) leaves the white-noise variance unchanged in every mode. Then $\sigma^2 = \hat{\sigma}_n^2 \sum_t w_t^2$. For self-noise, $\sigma^2 = \sum_t w_t^2\sigma_t^2$, with σ_t^2 the local mean square of the series rebuilt from the modes $k > N/4$, divided by 3/4.

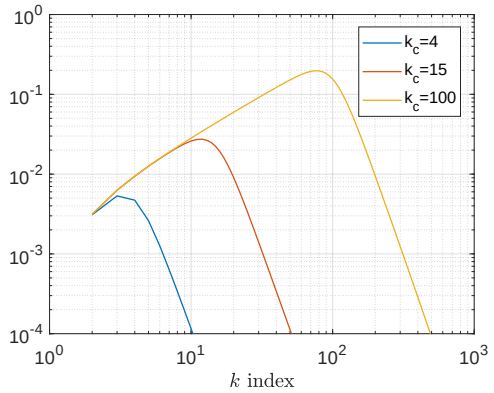


Figure 2. $\mathbf{\Lambda}^{1/2}_{D_1^T D_1}\mathbf{D}_{\rm filter}$ band pass filter response for $k_c = 4, 15, 100$ for an $O = 3$ filter defined by Eq. (21).

3. RESULT: NUMERICAL EXAMPLE

3.1. *Selection of Filter Parameters*

We exemplify the method discussed in §2.1 by demonstrating the selection of filter parameters to correctly recover the underlying $I(t)$ structure in the presence of noise. Fig. 3 (top) shows 1000 noisy $I = |X|^2 + |Y|^2$ data points with an underlying double pulse as shown, representing total power I calculated from perpendicular X and Y linearly polarized voltage components. The noise-free component of X consists of two Gaussian pulses delayed by 370 and 630 units with standard deviations of 80 and 85, respectively. The amplitude is normalized to the pulse peak. The noise-free amplitude of the Y component is taken as $0.5X$. The structure parameter of the noise-free pulse is $\|\mathbf{D}_1\mathbf{i}_{\rm noiseless}\| = 0.18$ as computed by the 2-norm of the first derivative. The noise in the complex voltages X, Y are independent and identically distributed Gaussian noise with zero mean and standard deviation $\sigma_{\Re(X)} = \sigma_{\Im(X)} = \sigma_{\Re(Y)} = \sigma_{\Im(Y)} = 0.4$. The DCT spectrum of the signal is depicted in Fig. 3 (bottom). It shows that the information occupies up to a k index of 10 to 20, beyond which the noise dominates. Therefore, we expect to select a low-pass filter cutoff between k_c of 10 to 20.

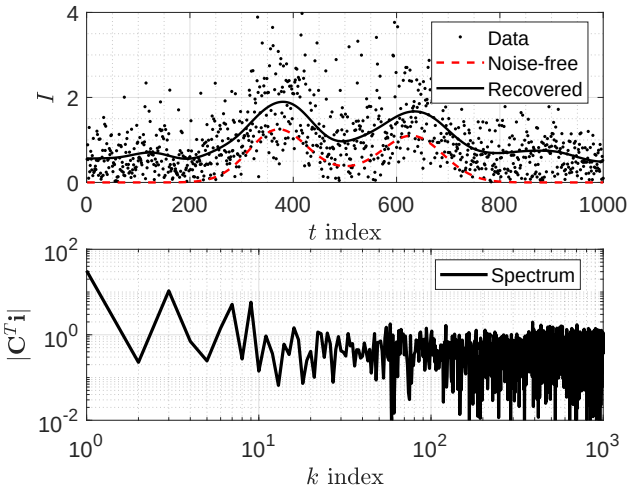


Figure 3. Example noisy pulse (top) and the corresponding spectrum in discrete cosine (bottom). The underlying pulse consists of two unity amplitude Gaussians. The solid black curve in the top plot represents the recovered intensity with filter parameters $O = 3$, $k_c = 15$.

The k_c estimate based on the DCT spectrum is confirmed by the calculated structure parameter and the uncertainty shown in Fig. 4 where the low-pass filter cutoff k_c was swept from 3 to 300 for filter order 2, 3, 4 in Eq. (21) to explore the effects of under- and over-filtering. The structure parameter was computed using Eq. (23) and the associated uncertainty was calculated using the original radius $\|\mathbf{D}_1\mathbf{S}\mathbf{\Delta}_\mathbf{i}\| / \left\|\mathbf{D}_1\mathbf{S}\tilde{\mathbf{i}}\right\|$. With Eq. (10) the uncertainties in Fig. 4 are lower by the square root of Eq. (11), and their k_c dependence needs recomputing. The structure parameter obtained from the smoothed noisy data matches the known noise-free value of 0.18 for k_c between 10 to 20, which agrees with that suggested by the DCT spectrum. This also coincides with the range of k_c values for which the uncertainty is minimized. We also note that the structure parameter becomes rapidly underestimated and overestimated as the k_c is underestimated (over-smoothing) and overestimated (under-smoothing), respectively. The uncertainty minima at k_c of approximately 5 for filter

orders 3 and 4 in Fig. 4 correspond to over-smoothing which may be readily identified from the DCT spectrum in Fig. 3 and confirmed by the recovered structure parameter. Proper determination of k_c is therefore critical, and we demonstrate that the DCT spectrum is a very effective tool for that purpose.

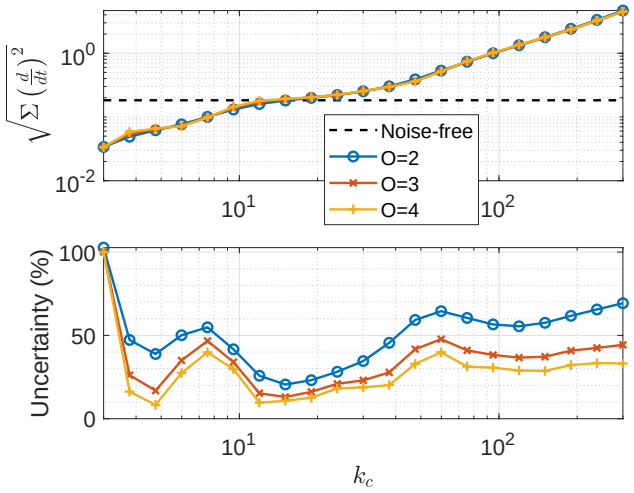


Figure 4. Recovered structure parameter (top) and uncertainty (bottom) of the structure parameter for k_c swept from 3 to 300 and filter order 2, 3, 4 in Eq. (21).

3.2. *Robustness with respect to Δ DM*

The previous sub-section operated on one-dimensional time-series data, which assumes that de-dispersion has already been performed. However, in reality, the dispersion measure is not known a priori (otherwise the problem would already be solved!) and hence it is necessary to consider whether de-dispersing across different trial DM values could affect the parameter choice and hence final results. In this sub-section, we simulate the effects of dispersion and de-dispersion on the double pulse already considered.

In the frequency domain, the net effect of dispersion and de-dispersion may be described as a multiplicative factor

$$e^{-j(k(\nu)-\tilde{k}(\nu))L} = e^{-j(2\pi)K_{\rm DM}\frac{-\Delta{\rm DM}[{\rm pc\,cm^{-3}}]}{\nu[{\rm MHz}]}}. \quad (26)$$

The units in Eq. (26) follow the convention described in Wilson et al. (2009); L is the distance to the source and $k(\nu)$ and $\tilde{k}(\nu)$ are the wavenumber and the wavenumber estimate, respectively; the difference between the actual DM and the DM estimate is $\Delta{\rm DM} = {\rm DM} - \tilde{\rm DM}$. Throughout, we define the delay constant $K_{\rm DM}$ to be $(1/0.241)\cdot 10^9 \approx 4.149\cdot 10^9\,{\rm MHz\,cm^3\,pc^{-1}}$. Hence, the effects of dispersion and de-dispersion may be simulated by multiplying Eq. (26) with the Fourier transform (FT) of the noisy complex voltages, followed by inverse FT of the product.

As an example, we took a 256×1024 matrix of complex white noise with $\sigma_{\rm re} = \sigma_{\rm im} = 1/\sqrt{2}$. This noise matrix was multiplied by two 2-D Gaussian real envelopes, then complex Gaussian noise with $\sigma_{\rm re} = \sigma_{\rm im} = 1/2\sqrt{2}$ was added. The result shown in Fig. 5 is the simulated data at $\Delta{\rm DM} = 0$. We chose the time delays between the components to mimic the 'sad trombone' effect, i.e. the frequency down-drift of burst components with time, often seen in repeating FRBs (e.g. Hessels et al. 2019).

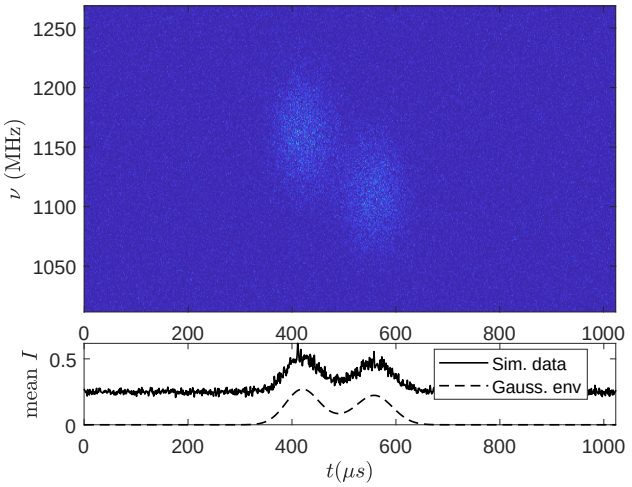


Figure 5. The frequency-time simulated frequency down-drifting data at $\Delta{\rm DM} = 0$ (top). The bottom figure shows the mean intensity over frequency of the noisy data (solid line) and the Gaussian envelope (dashed line).

We applied $\Delta{\rm DM}$ from -0.5 to $0.2\,{\rm pc\,cm^{-3}}$ in $0.01\,{\rm pc\,cm^{-3}}$ steps. The resulting noisy mean intensities vs. time over $\Delta{\rm DM}$ are shown in Fig. 6. The intensity of the noise-free data was simulated by applying the corresponding group delay $\tau_d = 4.149\times 10^3\Delta{\rm DM}[{\rm pc\,cm^{-3}}]/f^2_{\rm MHz}$ to the intensity envelope. To filter the noise in Fig. 6, we applied the same strategy discussed in §3.1. We performed DCT of the frequency-time 256×1024 data shown Fig. 5 and then visually inspected the resulting DCT spectrum (not shown). The transition between information and noise occurred at k index of approximately 35. Therefore, we set $k_c = 35$ and $O = 3$ was kept the same as the previous section.

Fig. 7 summarizes the robustness of the method we propose in this paper with respect to $\Delta{\rm DM}$. Fig. 7 (top) shows the comparison between the recovered (from the noisy data) and the known structure parameters (from the noise-free intensity envelope that was time shifted as per the group delay at each frequency). The dif-

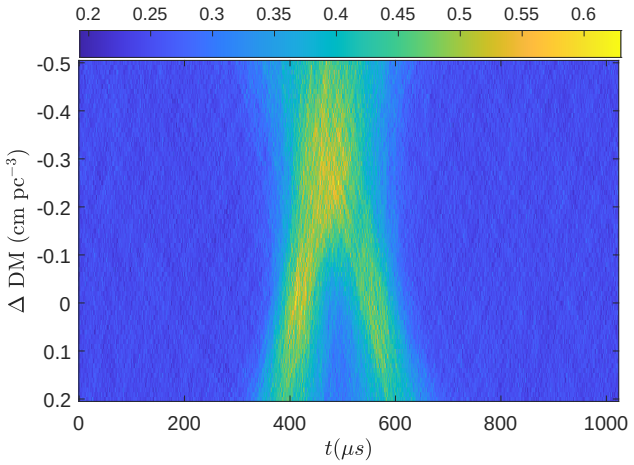


Figure 6. Intensities vs. time and ΔDM of simulated frequency down-drifting data before filtering.

ference between the two are plotted in the second row. The uncertainty computed using Eq. (10) the original radius is shown in the third row (needs recomputing with Eq. (10)). Finally, the maximum recovered intensity is shown at the bottom of Fig. 7. From this figure, we infer the following. The filter recovers the correct structure parameters to within a few percent for all trial ΔDMs. The recovered highest structure parameter at $\Delta\mathrm{DM} \approx 0$ and the maximum intensity at $\Delta\mathrm{DM} \approx -0.26\,\mathrm{pc\,cm^{-3}}$ are consistent with the noise-free intensity envelope. For the selected filter, the uncertainty of $\sim 5\%$ to 10% in the measured structure is comparable across ΔDMs. At $\Delta\mathrm{DM} = 0$, where the structure is maximised but the intensity is lower than the highest peak, the uncertainty appears slightly lower.

To derive an uncertainty in ΔDM, we take the noise estimates $\Delta_{\mathbf{i}}$ at the structure-maximised value ΔDM=0, and subtract the noise estimates at other values of ΔDM. This difference also contains the burst residual difference, which is not noise. This is shown in Figure 8. The magnitude of the relative errorsas per , from the original radius form of Eq. (15) (needs recomputing), are given in Fig. 7. For values of ΔDM near 0, the relative uncertainty (dashed line in third panel from top) is smaller than the absolute uncertainty (solid line) due to strong correlations in the filtered noise. At large values of ΔDM however, the noise becomes uncorrelated, and so is approximately $\sqrt{2}$ times the absolute value.

To derive error bounds for ΔDM , we add the error to the structure as per the right-hand-side The DM uncertainty is the scatter of the structure peak, σ_{DM} of Eq. (16), also shown in Fig. 7. The uncertainty in ΔDM corresponds to those regions where the relative noise plus the intrinsic structure exceeds that of the structure at ΔDM= 0, i.e. which can not be excluded by Eq. ((16)). Thus we find $\Delta\mathrm{DM}= 0^{+0.08}_{-0.04}$ (27). If an interval test such as Eq. (16) is used instead, the region where $\left\|\mathbf{D}_1\mathbf{Si}_{\hat{d}}\right\| - \|\mathbf{D}_1\mathbf{Si}_d\| \leq z\,\sigma_{12}$ has half-width $2z\,\sigma_{\mathrm{DM}}$, so $z = \frac{1}{2}$ gives 1σ (Appendix B). The original value here, from the radius with $z = 1$, needs recomputing; in our simulations that procedure overestimates σ_{DM} by a factor of 9–13.

4. RESULT: APPLICATION TO FRB DATA

The above methods can be applied to determine the correct DM of an FRB, and the uncertainty thereon. Dedispersing an FRB for many trial DMs produces voltage time series that can be converted into intensity time-series **i**. The procedure of Sec. §2 then produces a corresponding measure of FRB structure and its uncertainty for each and every trial DM. This then allows the trial DM that maximizes structure to be identified, and an error in DM assigned to include the range of trial DMs with structure consistent with the maximum value, given the calculated structure error.

4.1. *FRB181112*

We apply our method to FRB181112 (Cho et al. 2020) data averaged to $1\,\mu$s. Fig. 9 shows the raw intensity data as a function of dispersion measure, away from the nominal value of $589.265\,\mathrm{pc\,cm^{-3}}$ that produces the maximum signal to noise (S/N) ratio. The spectrum of the intensity time series was obtained by the multiplication with the discrete cosine matrix discussed in Sec. 2.2. This is shown in Fig. 10. We observe that the information occupies up to $k = 30$ index, after which it is dominated by noise. Based on this information, we selected the filter cutoff of $k_c = 30$ and chose a $O = 3$ filter as per Eq. (21).

The first derivative of the smoothed data is reported in Fig. 11. Finally, the structure parameter and the corresponding uncertainty are plotted in Fig. 12. We see that the highest structure coincides with ΔDM=0 which is the maximum S/N. The uncertainty has a trend that increases away from $\Delta\mathrm{DM} = 0$, while the relative uncertainty compared to $\Delta\mathrm{DM} = 0$ increases even more strongly. We find that allowing the structure to fluctuate upwards by one standard deviation of relative noise allows structure within $\Delta\mathrm{DM} = 0 \pm 0.011\,\mathrm{pc\,cm^{-3}}$ to exceed the peak value of 12.2×10^{-5} at $\Delta\mathrm{DM} = 0$, i.e. our uncertainty in ΔDM is $0.011\,\mathrm{pc\,cm^{-3}}$. This value used the original radius and needs recomputing with Eq. (27).

4.2. *FRB210117*

Next, we consider FRB210117 with time-domain data at $10\,\mu$s resolution shown in Fig. 13 for the indicated

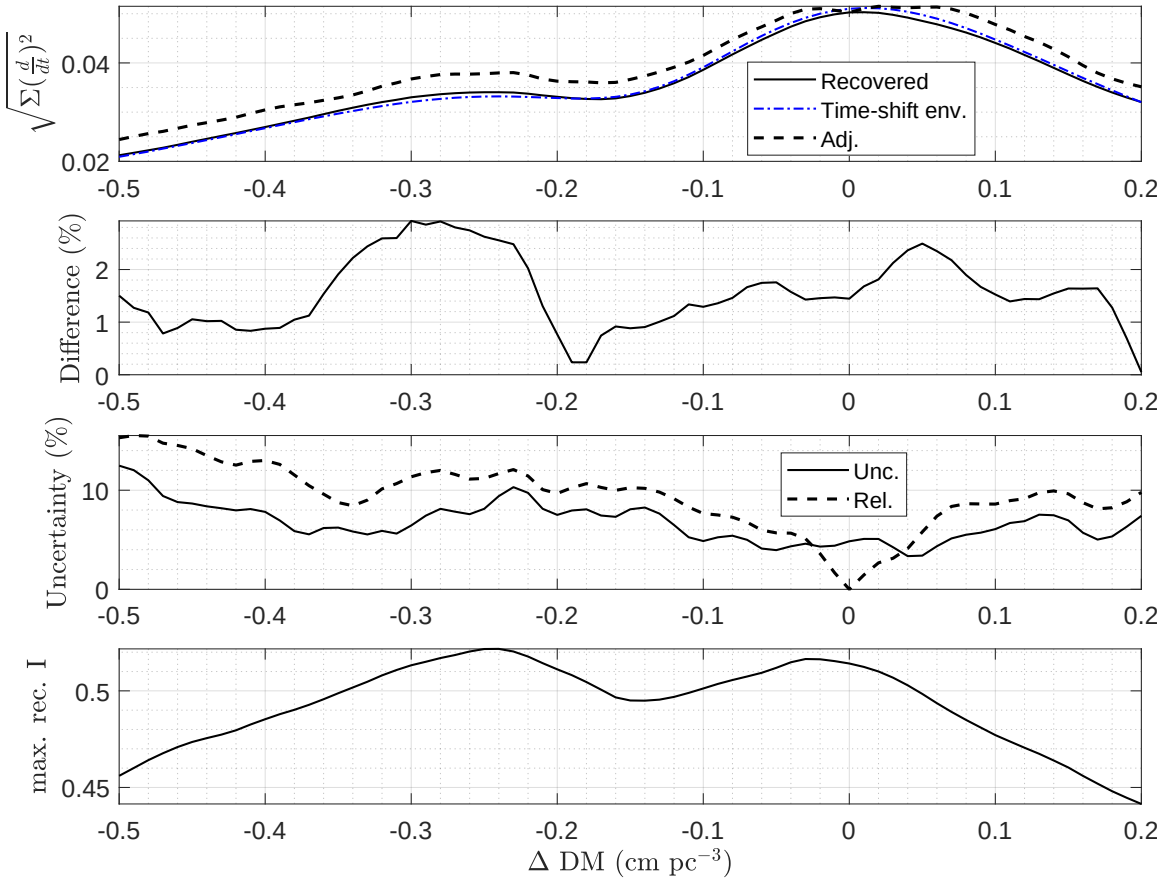


Figure 7. Calculated structure parameter (top), difference between the structure parameter of the recovered signal and the original time-delayed envelope (second row), uncertainty of the recovered structure parameter (third row), maximum recovered intensity (bottom), of simulated frequency down-drift data.

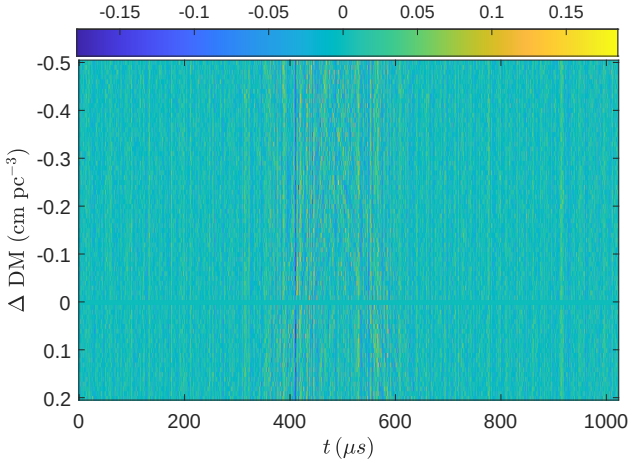


Figure 8. The vector $\mathbf{D}_1\mathbf{S}\,(\mathbf{\Delta}_{\Delta\mathrm{DM}=0} - \mathbf{\Delta}_{\Delta\mathrm{DM}})$ from used in the original radius form of Eq. (15), calculated for simulated frequency down-drifting data.

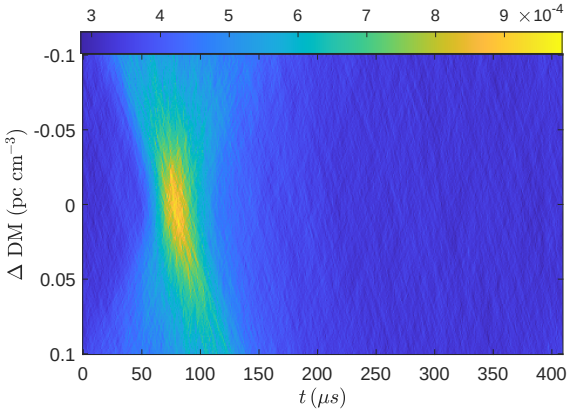


Figure 9. Intensity of FRB181112 vs. time and Δ DM in pc cm^{-3} from the DM that produces maximum S/N.

ΔDM, relative to the nominal S/N maximizing value of 729.2 pc cm^{-3} (Bhandari et al. 2022). This FRB appears to have a secondary peak for ΔDM < 0. The spectrum of this FRB (not shown) suggests a spectral transition at a k value between 25 and 35. This is shown in Fig. 14 where we plot the best-fitting ΔDM and 68% uncertainty bounds as per the original radius form of Eq. (16) (not 68%) over filter cutoff k_c (with $O = 3$). It shows that $k_c \approx$25–35 offers consistent values of the structure-

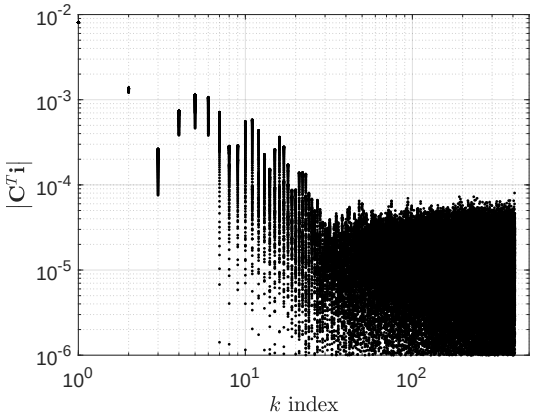


Figure 10. Spectrum of FRB 181112 in discrete cosine domain shown in absolute values for all ΔDM.

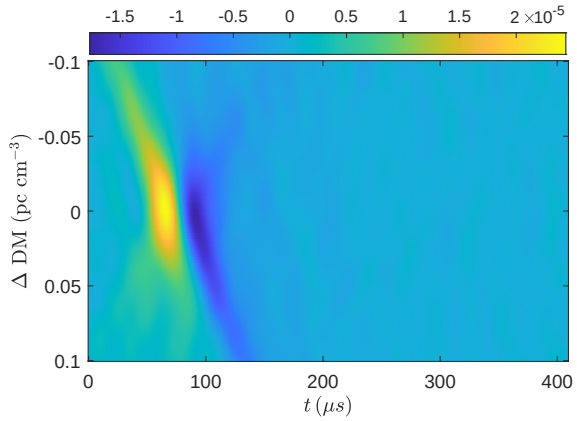


Figure 11. First derivative of the intensity of FRB181112 vs. time and ΔDM after smoothing using $k_c = 30$ and $O = 3$ filter.

maximising ΔDM with the narrowest error margin. Using lower values of k_c results in over-smoothing, and thus a loss of signal structure; while using higher values of k_c adds noise, increasing the random error in ΔDM.

The structure parameter and the associated uncertainty are plotted in Fig. 15. Using the same procedure as for FRB 20181112, we find a structure-maximising $\Delta\mathrm{DM} = -0.1^{+0.36}_{-0.23}\,\mathrm{pc\,cm^{-3}}$ (uncertainty to be recomputed with Eq. (27)). Thus, we cannot exclude that this FRB's apparent structure-maximising DM is significantly different from its signal-to-noise maximising DM.

5. CONCLUSION

We demonstrated that computing the structure parameter using the 2-norm, $\sqrt{\Sigma(d/dt)^2}$, permits a direct calculation of the uncertainty of the resulting structure parameter. The uncertainty of the first derivative of

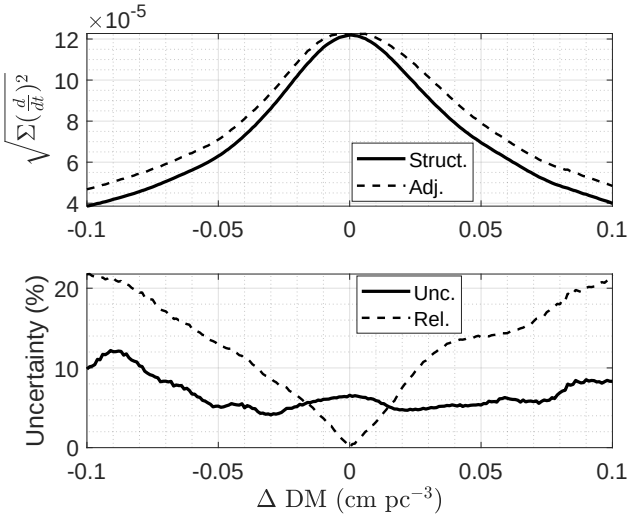


Figure 12. Solid lines: Structure parameter of FRB181112 computed using $\sqrt{\Sigma(d/dt)^2}$ ($k_c = 30, O = 3$) (top), and the corresponding uncertainty (bottom). Dashed lines: 68% upper limits from the original radius test (not 68%; Sect. 3.2) on structure (top), and relative error (bottom), with respect to ΔDM= 0 pc cm^{-3}.

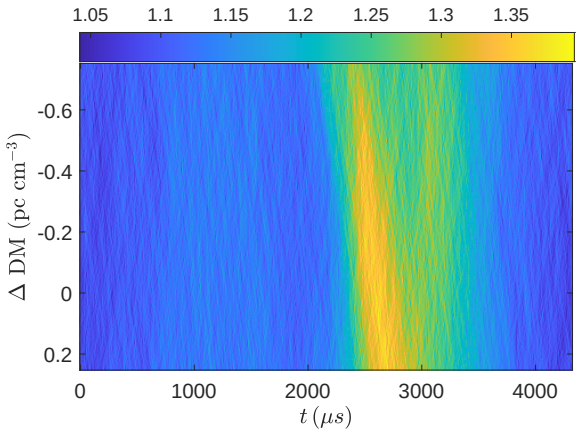


Figure 13. Intensity before smoothing of FRB210117 vs. time and ΔDM from the DM that produces maximum S/N.

the smoothed data is calculated by taking the difference between the structure parameter of the smoothed data and the structure parameter of the is the scatter of the noise projected onto the structure direction, Eq. (9). The length of the residual between the smoothed and "double-smoothed" data. The latter can be obtained easily by taking the smoothed data (before taking the derivative) and plugging it back as the input to the algorithm" data overestimates it by the square root of Eq. (11), a factor of 5–9 in our simulations. We have also

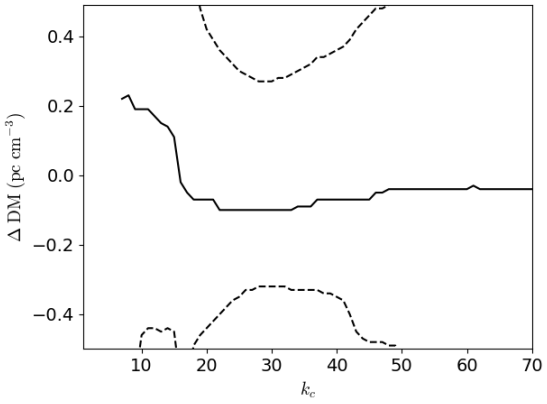


Figure 14. Structure maximising ΔDM (solid) and 68% error bounds from the original radius test (dashed; not 68%) for FRB210117 as a function of filter cutoff k_c ($O = 3$).

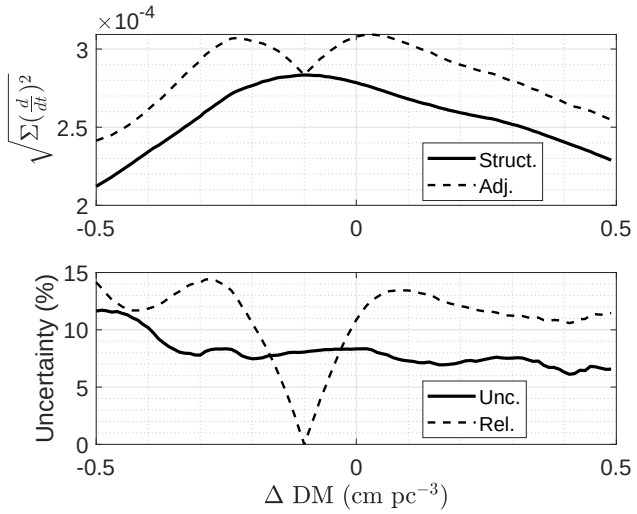


Figure 15. Solid lines: Structure parameter of FRB210117 (top) and the corresponding uncertainty with $k_c = 27$ and $O = 3$ smoothing filter (bottom). Dashed lines: 68% upper limit from the original radius test (not 68%; Sect. 3.2) on structure (top), and relative uncertainty (bottom), with respect to ΔDM= $-0.2\,\mathrm{pc\,cm^{-3}}$.

shown how to use this uncertainty for statistical tests for different intrinsic structure between two filtered signals.

This method was applied to FRB181112 and FRB210117. In the case of FRB181112, the DM that produces the highest S/N is the same as that with the highest structure within an uncertainty in ΔDM of $\pm 0.011\,\mathrm{pc\,cm^{-3}}$. As for FRB210117, the highest structure occurred at a DM = $729.1^{+0.36}_{-0.23}\,\mathrm{pc\,cm^{-3}}$, compared to the signal-to-noise maximising DM of $729.2\,\mathrm{pc\,cm^{-3}}$. Both DM uncertainties need recomputing with Eq. (27).

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APPENDIX

We now briefly discuss other definitions of structure parameters: $\Sigma\left(d/dt\right)^2$ (Hessels et al. 2019), $\Sigma\left|d/dt\right|$ (Gajjar et al. 2018), and $\Sigma\left(d/dt\right)^4$ (Josephy et al. 2019). We note that $\Sigma\left|d/dt\right|$ is the 1-norm of the vector derivative of the smoothed data (Strang 2019, see Ch. 1). This means the triangle inequality is fulfilled, $\|\mathbf{x}+\mathbf{y}\|_1 \leq \|\mathbf{x}\|_1 + \|\mathbf{y}\|_1$ and the upper bound is defined. However, the inner product is not defined in the 1-norm (Strang 2019, see. Ch. 1). As a result, the corresponding lower bound cannot be shown to be that in Eq. (12) for the 2-norm case[2]. Furthermore, we are not able to relate the standard deviation estimate, such as Eq. (7) in the 2-norm, to a corresponding statistical quantity in the 1-norm.

$\Sigma\left(d/dt\right)^2$ and $\Sigma\left(d/dt\right)^4$ do not conform to the norm definition because they are missing $\sqrt{}$ and $\sqrt[4]{}$ operators which would have made them 2-norm and 4-norm, respectively (Laub 2005, see Ch. 7). Therefore for these definitions, the triangle inequality does not apply and we are not able to establish the bounds of the summation of two vectors.

In Gajjar et al. (2018), the uncertainty of the structure parameter was inferred from the off-pulse system noise. This assumes that the noise during the FRB pulse is the same that of the system noise. However, it is entirely possible that the FRB pulse is noise-like and hence the on-pulse noise is different from the off-pulse noise. In

[2] One can use the reverse triangle inequality to show $\|\mathbf{x}-\mathbf{y}\|_1 \geq \big|\|\mathbf{x}\|_1 - \|\mathbf{y}\|_1\big|$, but this not the same as establishing the lower bound of $\|\mathbf{x}+\mathbf{y}\|_1$.

fact, the noise statistics of the FRB is in itself an interesting subject for study, and therefore, it is desirable that the uncertainty be inferred from that same on-pulse data. Measuring σ_n from the $k > N/4$ band of the on-pulse spectrum (Eq. 25) does this.

APPENDIX B: CORRECTED UNCERTAINTY

Expansion. With $\mathbf{a}$, $\mathbf{e}$, $\hat{\mathbf{a}}$, $e_\parallel$ and $\mathbf{e}_\perp$ as in §2.1, Pythagoras gives $\left\|\mathbf{D}_1\tilde{\mathbf{i}}\right\|^2 = (\|\mathbf{a}\| + e_\parallel)^2 + \|\mathbf{e}_\perp\|^2$. For $\|\mathbf{e}_\perp\| \ll \|\mathbf{a}\|$, $\sqrt{X^2+Y^2} \simeq X + Y^2/2X$ gives Eq. (8) to second order in the noise.

Variance. $\|\mathbf{a}\|$ is fixed and the last term of Eq. (8) is second order, so to first order the variance of the structure parameter is that of $e_\parallel = \mathbf{w}\mathbf{n} = \sum_t w_t n_t$. Since $\langle e_\parallel\rangle = \sum_t w_t \langle n_t\rangle = 0$, $\sigma^2 = \langle e_\parallel^2\rangle = \sum_t \sum_{t'} w_t w_{t'} \langle n_t n_{t'}\rangle = \mathbf{w}\,\mathrm{Cov}(\mathbf{n})\,\mathbf{w}^T$, which is Eq. (9). For noise uncorrelated between samples only $t' = t$ survives, $\sigma^2 = \sum_t w_t^2 \sigma_t^2$.

Estimated direction. Estimating $\hat{\mathbf{a}}$ by $\mathbf{D}_1\mathbf{S}\mathbf{i}/\|\mathbf{D}_1\mathbf{S}\mathbf{i}\| = (\mathbf{a}+\mathbf{e})/\|\mathbf{a}+\mathbf{e}\|$ tilts it by $\theta \simeq \|\mathbf{e}_\perp\|/\|\mathbf{a}\|$. For any fixed matrix $\mathbf{M}$ and noise uncorrelated between samples, $\langle\|\mathbf{M}\mathbf{n}\|^2\rangle = \sum_j \sum_t M_{jt}^2 \sigma_t^2$. With $\mathbf{M} = \mathbf{D}_1\mathbf{S}$ this gives $\langle\|\mathbf{e}\|^2\rangle \simeq 40$–$45\,\sigma^2$ in our simulations, and $\langle\|\mathbf{e}_\perp\|^2\rangle = \langle\|\mathbf{e}\|^2\rangle - \sigma^2$, so $\theta \simeq 6\sigma/\|\mathbf{a}\|$: the estimate is good when $\|\mathbf{a}\| \gg 6\sigma$.

Bias. $b = \langle\left\|\mathbf{D}_1\tilde{\mathbf{i}}\right\|\rangle - \|\mathbf{a}\| = \langle\|\mathbf{e}_\perp\|^2\rangle/2\|\mathbf{a}\| = (\langle\|\mathbf{e}\|^2\rangle - \sigma^2)/2\|\mathbf{a}\|$. It varies slowly with DM, does not move the peak at first order, and is not an uncertainty.

What Eq. (7) measures. The cross term of $\|\mathbf{d}_s + \mathbf{d}_n\|^2$ has zero mean, so $\langle\|\mathbf{D}_1\mathbf{S}\boldsymbol{\Delta}_\mathbf{i}\|^2\rangle = \|\mathbf{d}_s\|^2 + \langle\|\mathbf{d}_n\|^2\rangle$, with $\langle\|\mathbf{d}_n\|^2\rangle$ from the expression above for $\mathbf{M} = \mathbf{D}_1\mathbf{S} - \mathbf{D}_1\mathbf{S}\mathbf{S}$, since $\mathbf{d}_n = (\mathbf{D}_1\mathbf{S} - \mathbf{D}_1\mathbf{S}\mathbf{S})\mathbf{n}$. Dividing by σ^2 gives Eq. (11). Per DCT mode k (Eq. 22), $\mathbf{D}_1\mathbf{S}$ multiplies by $\mathrm{eig}_k^{1/2} f(k)$ and $\mathbf{D}_1\mathbf{S} - \mathbf{D}_1\mathbf{S}\mathbf{S}$ by $\mathrm{eig}_k^{1/2} f(k)[1 - f(k)]$, which peaks near k_c, where the filter passes half. Only the residual noise near k_c survives, so the first factor of Eq. (11) is $\langle\|\mathbf{e}\|^2\rangle/\sigma^2 \simeq 40$–$45$ reduced by $\langle\|\mathbf{e}\|^2\rangle/\langle\|\mathbf{d}_n\|^2\rangle \simeq 7.7$, i.e. 5–6. $\mathbf{d}_s$ also lies near k_c, where the cutoff is chosen so that the burst is comparable to the noise; it is not noise, and $\|\mathbf{d}_s\|^2/\langle\|\mathbf{d}_n\|^2\rangle \propto (\mathrm{S/N})^2$ at fixed k_c.

DM uncertainty. At trial DM d, the structure parameter is $\mathcal{S}_d = \left\|\mathbf{D}_1\tilde{\mathbf{i}}_d\right\| \simeq \left\|\mathbf{a}(d)\right\| + e_\parallel(d)$. Let d_0 be the peak of the noise-free $\left\|\mathbf{a}(d)\right\|$. Near it, $\left\|\mathbf{a}(d)\right\| \simeq \left\|\mathbf{a}(d_0)\right\| + \frac{1}{2}\|\mathbf{a}\|''(d_0)(d-d_0)^2$ and $e_\parallel(d) \simeq e_\parallel(d_0) + e_\parallel'(d_0)(d-d_0)$: the noise value lifts the curve and the noise slope tilts it. Setting the derivative to zero at the measured peak $\hat{d}$ gives $\hat{d} - d_0 = -e_\parallel'(d_0)/\|\mathbf{a}\|''(d_0)$, so $\sigma_{\mathrm{DM}} = \mathrm{std}[e_\parallel'(d_0)]/|\|\mathbf{a}\|''(d_0)|$. Write $e_\parallel(d) = \sum_t w(t;d)\,n(t;d)$, with $w(t;d)$ the weight row at DM d and $n(t;d) = \sum_\nu x_\nu(t + d\,\tau_\nu)$ the dedispersed noise, where $x_\nu(t)$ is the noise in channel ν at sample t and $\tau_\nu = K_{\mathrm{DM}}(\nu^{-2} - \nu_{\mathrm{ref}}^{-2})/t_{\mathrm{samp}}$ is its delay per unit DM in samples. Relabelling $t \to t - d\,\tau_\nu$ in each channel moves all the dependence on d into the smooth weights, $e_\parallel(d) = \sum_\nu \sum_t w(t - d\,\tau_\nu; d)\,x_\nu(t)$, so $e_\parallel'(d_0) = \sum_\nu \sum_t w_\nu'(t)\,x_\nu(t)$ with $w_\nu'(t) = \partial_d\, w(t - d\,\tau_\nu; d)\big|_{d_0}$. For channel noise independent between channels and samples, with variance σ_ν^2,

$$\sigma_{\mathrm{DM}} = \frac{\left(\sum_\nu \sigma_\nu^2 \sum_t w_\nu'(t)^2\right)^{1/2}}{|\|\mathbf{a}\|''(d_0)|}. \tag{27}$$

This holds for any burst morphology (drift, scattering) and does not depend on ν_{ref}. In practice $d_0 \to \hat{d}$, and w_ν' is a central difference,

$$w_\nu'(t) \simeq \frac{1}{2\delta}\big[w(t - (\hat{d}+\delta)\tau_\nu; \hat{d}+\delta) - w(t - (\hat{d}-\delta)\tau_\nu; \hat{d}-\delta)\big],$$

with each row computed from the data dedispersed at that DM and $\delta \simeq \sigma_{\mathrm{DM}}/2$. $\|\mathbf{a}\|''$ is the curvature of the measured structure curve: either $2c_2$ from a least-squares parabola $c_0 + c_1 d + c_2 d^2$ over $|d - \hat{d}| \lesssim 2$–$3\,\sigma_{\mathrm{DM}}$, or the second difference $[\mathcal{S}_{\hat{d}+h} - 2\mathcal{S}_{\hat{d}} + \mathcal{S}_{\hat{d}-h}]/h^2$ with $h \simeq 2\sigma_{\mathrm{DM}}$, iterating from an initial guess of σ_{DM}; narrower windows pick up the curvature of the noise. Neither assumes the same profile in every channel. Alternatively, pass $J \sim 100$–200 off-pulse noise segments through the same dedispersion and take σ_{DM} as the scatter over j of the peak of $\mathcal{S}_d + \sum_t w(t;d)\,n^{(j)}(t;d)$.

Interval tests. With $\Delta d = d - \hat{d}$, near the peak $\mathcal{S}_{\hat{d}} - \mathcal{S}_d \simeq \frac{1}{2}|\|\mathbf{a}\|''|\Delta d^2$, and the noise of the difference is $e_\parallel'\Delta d$, with standard deviation $\mathrm{std}[e_\parallel']|\Delta d|$. So $\{d : \mathcal{S}_{\hat{d}} - \mathcal{S}_d \le z\,\mathrm{std}[e_\parallel']|\Delta d|\}$ has half-width $2z\sigma_{\mathrm{DM}}$.

Numerical check. Toy model with a two-component burst, 84 channels, $N = 2048$, white noise and known DM: Monte-Carlo scatter of the structure parameter 0.240, 0.241 (burst amplitude 0.5, 1.0), Eq. (9) 0.252, 0.256, original radius 1.17, 2.21; square root of Eq. (11) 4.6, 8.6 against measured 4.9, 9.2; original DM half-width 8.7 and 13 times the Monte-Carlo scatter of $\hat{d}$. For Eq. (27), a 32-channel toy with a drifting two-component burst gives 0.97 ± 0.03 of the first-order Monte-Carlo scatter of $\hat{d}$ and 0.91–0.96 of the full search at S/N 14–38 (the remainder is second order); estimated from each realisation's data alone it gives a median of 0.98 (16–84%: 0.89–1.12) and 0.94 (0.90–0.99) of the Monte Carlo at S/N 14 and 38; at S/N 14, noise in the data-built weight rows partly offsets the second-order shortfall.

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